

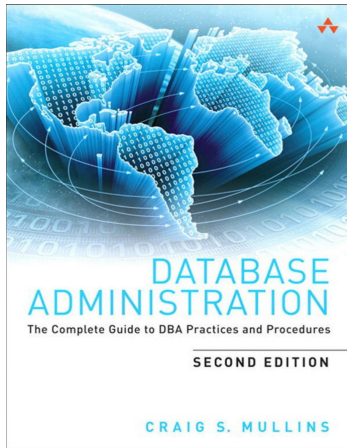
Database Administration

Lecture 10: Database Security.

Mullins, 2013.

20 de abril de 2026

Database Administration: Database Security.



Content has been extracted from *Database Administration: The Complete Guide to DBA Practices and Procedure (Chapter 14)*, by Craig Mullins, 2013. Visit <https://www.oreilly.com/library/view/database-administration-the/9780133012743/>.

Plan

Overview

Data Breaches

Database Security Basics

Granting and Revoking Authority

Authorization Roles and Groups

Other Database Security Mechanisms

Encryption

SQL Injection

Auditing

External Security

Overview

- ▶ Database security is crucial due to increasing data breaches and regulations.
- ▶ Data breaches are costly and prevalent.
- ▶ Databases are a prime target for hackers.
- ▶ Securing data requires a comprehensive plan.

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Data Breaches

- ▶ Data breaches continue to rise, with significant impact.
- ▶ Over 544 million records were breached between 2005 and 2012.
- ▶ Average cost per lost record is between \$90 and \$305.
- ▶ Costs include legal fees, customer losses, and bad publicity.
- ▶ Database servers are the main source of breached data.

Data Breaches: Impact and Trends (2021–2025)¹

- ▶ **Global Surge:** Data breaches reached record levels, with over **2.5 billion records** compromised in the last five years.
- ▶ **Financial Impact:**
 - ▶ Average cost per lost record: **\$173**.
 - ▶ High-risk sectors (e.g., Healthcare): Up to **\$429** per record.
- ▶ **Primary Vulnerabilities:** Cloud repositories and misconfigured database servers account for **45 %** of all incidents.
- ▶ **Total Costs:** Includes forensic investigations, legal fees, and regulatory fines often exceeding \$250,000.

¹IBM / Ponemon Institute: The “Cost of a Data Breach Report” (2025 and 2024 editions).

Data Breaches: Impact and Trends (2021–2025) (Cont.)

Metric	2005–2012 (Legacy)	2021–2025 (Updated)
Total Records	544 Million	> 2.5 Billion
Cost per Record	\$90 – \$305	\$173 – \$429
Main Source	Database Servers	Cloud & Misconfigurations

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Database Security Basics

- ▶ DBMS controls all database resources.
- ▶ Users must be granted privileges to perform operations.
- ▶ DBA is typically responsible for database security.
- ▶ Key aspects of database security:
 - ▶ Authentication (Who is it?)
 - ▶ Authorization (Who can do it?)
 - ▶ Encryption (Who can see it?)
 - ▶ Audit (Who did it?)

Authentication

- ▶ Strong authentication is essential.
- ▶ Logins (user IDs) with passwords are required.
- ▶ Information needed for login creation:
 - ▶ Password
 - ▶ Default database
 - ▶ Default language
 - ▶ User's full name
 - ▶ Additional details (e.g., email, phone number)
- ▶ Passwords should be changed regularly.

Password Guidance

- ▶ Passwords should be difficult to guess.
- ▶ Guidelines:
 - ▶ Minimum 6 characters, longer if possible.
 - ▶ Use a combination of alphabetic and numeric characters.
 - ▶ Avoid complete words.
 - ▶ Do not embed personal information.
 - ▶ Consider concatenating unrelated words.
 - ▶ Use mnemonic devices, but not obvious ones.
 - ▶ Avoid common and weak password archetypes.
- ▶ Work with security administration to create and distribute guidelines.

Open Source Tools: DBMS Enforcement

- ▶ **PostgreSQL: credcheck**
 - ▶ Enforces complexity (length, digits, special characters).
 - ▶ Rejects passwords containing the username.
 - ▶ Prevents plaintext password exposure in logs.
- ▶ **MySQL/MariaDB: validate_password**
 - ▶ **Levels:** Low (length), Medium (complexity), Strong (dictionary).
 - ▶ Dictionary checks prevent common/weak words.
 - ▶ Provides a “strength score” for proposed credentials.

Open Source Tools: Automation & Identity

- ▶ **OpenBao (Secrets Management²)**
 - ▶ **Dynamic Secrets:** can generate secrets on-demand for some systems, such as Kubernetes or SQL databases.
 - ▶ **Identity based access:** Use a unified ACL system to broker access to systems and secrets and merges identities across providers.
 - ▶ Provides encryption as a service with centralized key management to simplify encrypting data.
- ▶ **Authentik / Keycloak (Centralized Identity)**
 - ▶ Centralizes “Login Administration” across multiple DBs.
 - ▶ Enforces Multi-Factor Authentication (MFA) and SSO.
 - ▶ Simplifies the “Drop Login” process when users leave.

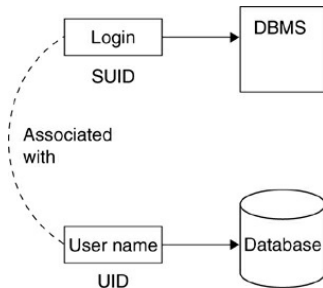
²HashiCorp Vault fork.

Login Administration

- ▶ Drop logins when no longer needed.
- ▶ Limit database users who can create database objects to DBAs only.
- ▶ Lock logins that may need to be reactivated.
- ▶ Drop logins that will never need to be reactivated.

Database Users

- ▶ Some systems require an additional account to use specific databases.
- ▶ Login (SUID): Accesses the DBMS or database server.
- ▶ User name (Database UID): Associated with the login account, required to access each database.
- ▶ Guest usage can be permitted by configuring a special user name.



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Granting and Revoking Authority

- ▶ DBA controls security using Data Control Language (DCL).
- ▶ DCL statements:
 - ▶ GRANT: Assigns permission to a database user.
 - ▶ REVOKE: Removes permission from a database user.
- ▶ GRANT statement requires a list of privileges and users.
- ▶ WITH GRANT OPTION allows a user to pass the authority to grant privileges.
- ▶ Avoid issuing GRANT and REVOKE statements from within an application program.
- ▶ Some DBMSs provide additional functionality (e.g., DENY in SQL Server).

Types of Privileges

- ▶ Basic types: access data, create objects, perform system functions.
- ▶ Common privilege types:
 - ▶ Table: Access and modify data within tables.
 - ▶ Database object: Create and drop database objects.
 - ▶ System: Perform system-wide activities.
 - ▶ Program: Create, modify, and use database programs.
 - ▶ Stored procedure: Execute specific functions and stored procedures.

Granting Table Privileges

- ▶ Control access to tables, views, and columns.
- ▶ Privileges:
 - ▶ SELECT: Enable user to select from table/view.
 - ▶ INSERT: Enable user to insert rows.
 - ▶ UPDATE: Enable user to update rows/columns.
 - ▶ DELETE: Enable user to delete rows.
 - ▶ ALL: Enable user to select, insert, update, and delete.
- ▶ Column-level privileges can be specified for SELECT and UPDATE.
- ▶ DBA typically grants table privileges in test environments.
- ▶ Production access should be controlled using program and stored procedure privileges.

Granting Database Object Privileges

- ▶ Control permissions to create database structures.
- ▶ Privileges depend on the DBMS and supported object types.
- ▶ Options to grant CREATE privileges on various objects (e.g., databases, tablespaces, tables, indexes).
- ▶ Ability to create objects is usually reserved for DBAs.
- ▶ Uncontrolled object creation can lead to difficulties in tracking and managing database objects.

Granting System Privileges

- ▶ Control which users can use DBMS features and commands.
- ▶ Privileges vary from DBMS to DBMS.
- ▶ Examples: archive logs, shut down/restart server, start traces, manage storage and caches.
- ▶ System privileges are granted system-wide, not at the database level.
- ▶ Should be granted with care, usually reserved for DBA and SA.

Granting Program and Procedure Privileges

- ▶ EXECUTE privilege allows users to execute programs or stored procedures.
- ▶ Granting privileges on programs/procedures is easier to control than on individual tables/columns.
- ▶ Procedural logic controls which tables and columns can be modified.
- ▶ DBA can better maintain data integrity by controlling changes programmatically.

Granting to PUBLIC

- ▶ Granting authorization to PUBLIC allows anyone who can log in to the DBMS to have that authority.
- ▶ Grants to PUBLIC cannot be given with the WITH GRANT OPTION.
- ▶ Using PUBLIC can appear easier but requires caution.
- ▶ DBA loses control over the object or resource when a privilege is granted to PUBLIC.
- ▶ Reserve PUBLIC usage for objects/resources that should be available to everyone.
- ▶ PUBLIC authority can be a shortcut if another security mechanism is in place (e.g., transaction processor security).

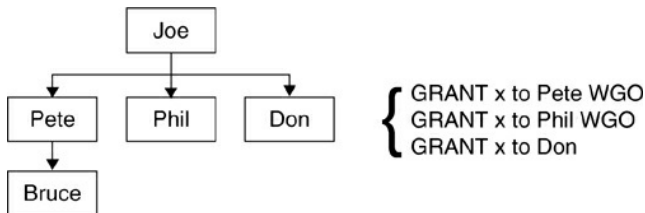
PUBLIC Authority and the System Catalog

- ▶ Dilemma: Whether to grant PUBLIC access to system catalog tables.
- ▶ Metadata in the system catalog is useful for DBAs, developers, and analysts.
- ▶ System catalog may contain sensitive information exploitable by hackers (e.g., authorization details, user IDs, login information).
- ▶ Exposes table names, making it easier for hackers to attempt unauthorized access or SQL injection.
- ▶ Refrain from granting blanket PUBLIC access to the system catalog tables.

Revoking Privileges

- ▶ REVOKE statement removes previously granted privileges.
- ▶ Syntax is the “flip side” of the GRANT syntax.
- ▶ Privileges are automatically revoked when a database object is dropped.
- ▶ Revoking a PUBLIC privilege does not remove it from users granted it in a separate GRANT statement.

Cascading REVOKEs



- ▶ When a revoke causes the DBMS to revoke additional related privileges.
- ▶ Example: Joe grants privilege X to Pete and Phil (with GRANT option), Pete grants X to Bruce, Joe grants X to Don (without GRANT option).
- ▶ If X is revoked from Joe, it will also be revoked from Pete, Phil, Don, and Bruce (cascading effect).
- ▶ To minimize impact, avoid granting privileges using the WITH GRANT OPTION.
- ▶ Some DBMS products allow revoking without cascading.

Chronology and Revokes

- ▶ Timing of GRANT or REVOKE can affect impact.
- ▶ Example: Grant DELETE on titles to public; COMMIT; REVOKE DELETE on titles from userx;
- ▶ In some DBMSs (e.g., Microsoft SQL Server), this sequence bars userx from deleting, even though PUBLIC was granted DELETE.
- ▶ Some DBMSs (e.g., DB2) do not permit such exclusions; PUBLIC overrides revokes.
- ▶ DBA must understand how GRANTs and REVOKEs work for each DBMS.

Label-Based Access Control (LBAC)

- ▶ Need for low-level access control is critical as systems become more sophisticated.
- ▶ LBAC secures each piece of data, ensuring only authorized users can perform authorized functions.
- ▶ Supports applications needing a more granular security scheme.
- ▶ Can specify who can read and modify data in individual rows and/or columns.

Homework

More in Lab 10...

Security Reporting

- ▶ DBA needs to monitor and report on user privileges.
- ▶ Database security is maintained in the system catalog.
- ▶ SQL can be used to retrieve information from system catalog tables.
- ▶ Some DBMSs provide views and system-stored procedures for easier retrieval.
- ▶ Protect the security of the system catalog, especially in production.

Security Reviews

- ▶ User security requirements evolve over time.
- ▶ Database security needs to change as new applications are added and business requirements change.
- ▶ Regular security reviews are necessary to ensure implemented security matches current requirements.
- ▶ Reports from system catalog tables can provide input for reviews.

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Authorization Roles and Groups

- ▶ DBMS may allow assigning users specific privileges to a role or built-in groups of privileges.
- ▶ Terminology varies among DBMSs (roles vs. groups).
- ▶ DBA needs to understand how each DBMS implements roles and groups to simplify security administration.

Roles

- ▶ Role: Collection of privileges.
- ▶ DBA creates a role and assigns privileges to it.
- ▶ Role can be assigned to one or more users, simplifying security administration.
- ▶ Example:

```
CREATE role MANAGER; COMMIT;  
GRANT select, insert, update, delete on employee to MANAGER;  
GRANT select, insert, update, delete on job_title to MANAGER;  
GRANT execute on payroll to MANAGER; COMMIT;  
GRANT MANAGER to user1; COMMIT;
```

- ▶ Additional users can be assigned the MANAGER role without needing to issue individual GRANT statements.

Groups

- ▶ Group-level authority is similar to roles.
- ▶ DBMS provides built-in groups that cannot be changed.
- ▶ Each DBMS implements group-level security differently with different group names and privileges.
- ▶ Similarities exist across DBMSs.

Common Groups

- ▶ System administrator (SA or SYSADM): Most powerful, can execute all commands and access all objects.
- ▶ Database administrator (DBADM or DBA): All privileges over a specific database, can access but not modify data.
- ▶ Database maintenance (DBMAINT): Privileges for maintaining database objects (e.g., utilities and commands).
- ▶ Security administrator: Privileges for granting/revoking security, login/password administration, auditing, security configuration.
- ▶ Operations control (OPER or SYSOPR): Authority to perform operational tasks like backup/recovery and terminating tasks.

Group Management

- ▶ Limit the number of users with SA role or group-level authority.
- ▶ SA capabilities are very powerful; only corporate DBAs and systems programmers should have this authority.
- ▶ Group-Level Security and Cascading REVOKEs:
 - ▶ Some group-level privileges allow users to grant privileges to others.
 - ▶ Revoking group-level authority from such a user also revokes any privileges they granted.
 - ▶ Similar to cascading REVOKEs with the WITH GRANT option.

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Using Views for Security

- ▶ Views can simplify some aspects of database security.
- ▶ Example: Employee table with sensitive information (salary, etc.).
- ▶ Granting SELECT privilege on the table can cause a security problem.
- ▶ A view can be created to omit sensitive information.
- ▶ Users can be granted SELECT on the view to access appropriate information.

Vertical Restriction with Views

- ▶ Example View:

```
CREATE VIEW emp_all AS
SELECT
    first_name, last_name, middle_initial,
    street_address, state, zip_code
FROM
    employee;
```

- ▶ Only information specified in the view can be retrieved.
- ▶ Eliminating columns from a base table to create a view is vertical restriction.
- ▶ Definition of sensitive data varies by organization.

Stored Procedures for Security

- ▶ Stored procedures enhance security by restricting direct access to base tables.
- ▶ Execution privilege must be explicitly granted or revoked.
- ▶ Ensures controlled data access through procedural logic.

Logic-Oriented Security

- ▶ Implement security based on conditions such as user identity and time of access.
- ▶ Prevent unauthorized data manipulation.
- ▶ Example: Restrict access to specific hours.

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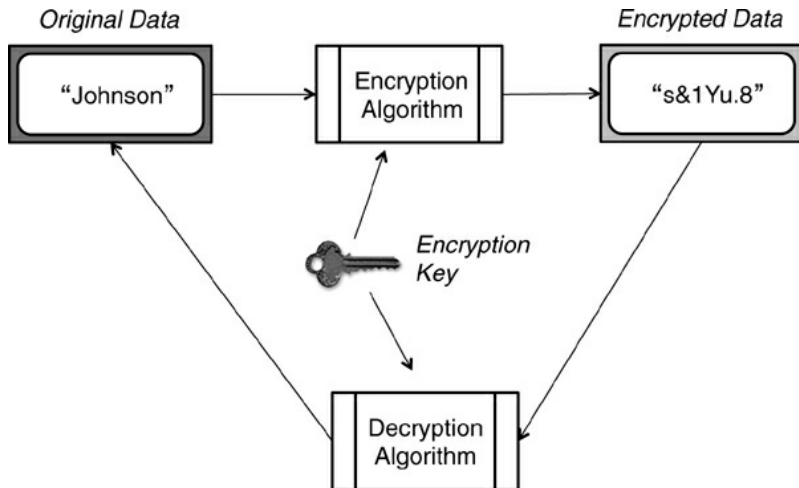
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Encryption Overview

- ▶ Protects data by transforming it into unreadable formats without decryption keys.
- ▶ Two primary types: **Data at Rest** and **Data in Transit** encryption.
- ▶ Used to comply with security regulations and prevent unauthorized access.

Encryption and Decryption



Types of Encryption

- ▶ Data at Rest: Protects stored data from unauthorized access.
- ▶ Data in Transit: Ensures secure data transfer over networks.
- ▶ Common encryption techniques: AES, DES, and SHA hashing.

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SQL Injection Attacks

- ▶ SQLi is a technique used by attackers to execute unauthorized SQL commands.
- ▶ Exploits improper input validation in web applications.
- ▶ Can lead to data leaks, unauthorized modifications, or even database destruction.

Preventing SQL Injection

- ▶ Validate all user inputs.
- ▶ Use prepared statements and parameterized queries.
- ▶ Restrict database permissions to limit exposure.
- ▶ Avoid concatenating user input in SQL statements.

Examples of SQL Injection Attacks

- ▶ In 2006, Russian hackers stole credit card data from a Rhode Island government website.
- ▶ In 2007, a hacker defaced Microsoft's UK website using SQL injection.
- ▶ In 2008, tens of thousands of PCs were infected via an automated SQL injection attack targeting Microsoft SQL Server.
- ▶ In 2010, a Swedish voter attempted an SQL injection attack in a write-in vote.
- ▶ In 2011, the Public Broadcasting System (PBS) was hacked using SQL injection.
- ▶ Other victims: Barracuda Networks, MySQL.com, Lady Gaga's website, Nokia, and Canon.

Recent SQLi Incidents (2023–2026)

- ▶ **MOVEit Transfer (2023)**: CVE³-2023-34362 allowed the exfiltration of data from **2,700+ organizations** affecting **93.3 million individuals**, including global firms and government agencies.
- ▶ **101news CMS Vulnerabilities (2025)**: CVE-2025-1871 allowed unauthenticated remote command execution via category parameters.
- ▶ **Driver Risks (2025)**: CVE-2025-1094 in PostgreSQL's `libpq` highlights risks in foundational database libraries.

³Common Vulnerabilities and Exposures

Case Study: Cybersecurity in Colombia

- ▶ **2023 State Attacks:** **32 state entities** (e.g., MinSalud, SIC) faced major breaches compromising citizen data.
- ▶ **Targeted Region:** In 2024, Colombia suffered **36 billion** cyberattack attempts, ranking 2nd in Latin America.
- ▶ **Improved Defense:** Reports for 2025 show a **48 % reduction** in successful incidents due to ColCERT strategies.
- ▶ **Sector Focus:** SQLi remains a top threat for the Colombian financial and healthcare sectors in 2026.

Modern Trends and Challenges

- ▶ **AI-Assisted Coding Risks:** AI tools often generate code lacking proper input validation, re-introducing SQLi flaws.
- ▶ **Technical Debt:** Mission-critical legacy systems (10–20 years old) often use outdated, vulnerable query patterns.
- ▶ **Persistence:** As of 2025, **38 %** of application weaknesses are still attributed to SQLi or Cross-Site Scripting (XSS).
- ▶ **DBA Role:** Moving from manual sanitization to automated policy enforcement using modern middleware.

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Database Auditing

- ▶ Tracks database activities to detect unauthorized access.
- ▶ Helps ensure compliance with security policies.
- ▶ Logs who accessed data, what was changed, and when.

Threats to Security

- ▶ Internal threats from disgruntled employees.
- ▶ External cyber-attacks targeting sensitive information.
- ▶ Auditing helps in identifying and mitigating such threats.

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External Security Mechanisms

- ▶ Protect database files at the operating system level.
- ▶ Use firewalls, access controls, and encryption.
- ▶ Prevent unauthorized access outside DBMS control.



Top 5 Fundamental Takeaways: Database Security

- 5 **Secure the “Four Pillars”:** Authentication, Authorization, Encryption, and Auditing. The DBMS centralizes control over all security resources.

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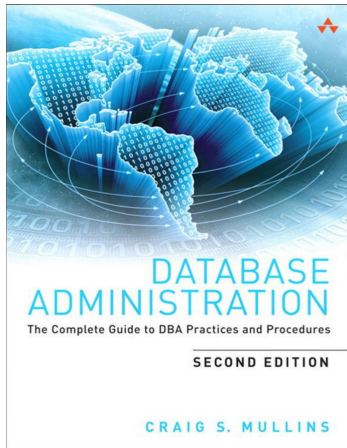
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- 2 **Granular Access Control:** Implement LBAC and Views for row/column-level security. Use Stored Procedures to enforce controlled modification logic.
- 1 **Proactive Defense:** Prevent SQLi using prepared statements and input validation. Perform regular reviews and auditing to detect evolving threats.

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